STM-GARSOS Processing: Concise Overview

STM-GARSOS workflow

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Objective and model

- Represent each dynamic image as $X_t(x, y) = \sum_{\ell=1}^{L} \mathrm{ST}_{t,\ell}(x, y) C_\ell(x, y)$.
- Estimate L spatial coefficient images from multicoil radial data.
- Synthesize the complete dynamic image series after the coefficient solve.

Raw-data preparation

- Load GAR-SOS metadata, raw k-space, and a vendor-reconstructed reference volume.
- Correct readout delay and prepare partition data.
- Estimate/apply coil-noise whitening.
- Generate normalized [kx, ky, kz] trajectory coordinates.
- Compute radial density-compensation weights.

Frame formation

- Flattened arrays retain one row per readout/partition/spoke sample.
- Spokes are grouped consecutively into temporal frames.
- Historical baseline: 15 spokes per frame and approximately 133 complete frames.
- Incomplete trailing spokes are discarded.

Calibration-data path

- Each frame is reconstructed with 3-D Kaiser-Bessel gridding.
- Oversampled grid: approximately 308 x 308 x 132.
- 3-D inverse FFT and centered resize produce a 224 x 224 x 96 volume.
- A selected z slice is transformed to 2-D Cartesian k-space.
- A central 32 x 32 calibration square is extracted.

STM calibration

- k_{Cal} is supplied to `STM_computation`.
- The calibration data span the temporal frames.
- The STM algorithm estimates ST maps and eigenvalues.
- Historical baseline: L = 7 and 224 x 224 spatial map dimensions.

Radial STM data

- The historical baseline used a direct centered 58-point partition IFFT.
- One z index is selected and the coil dimension is preserved.
- Output radial data: [readout x spokes/frame x coils x frames].
- In-plane kx/ky coordinates are retained for the NUFFT.

Multicoil radial operator

- Forward: $d_c = \mathcal{N}_{\mathrm{radial}}(S_c \odot X)$.
- Adjoint: $\mathcal{F}^H(d) = \sum_c S_c^* \odot \mathcal{N}_{\mathrm{radial}}^H(d_c)$.
- Sensitivity maps are forward receive-coil maps.
- Coils remain separate in the data fidelity term.

STM operator and PCG

- STM synthesis maps coefficients C to dynamic images X.
- The radial operator maps X to predicted multicoil radial data.
- Solve $(\mathcal{A}^H\mathcal{A})C = \mathcal{A}^H y$ with PCG.
- Optional Tikhonov solve adds lambda I.

Output and historical caveats

- Output: coefficient images plus synthesized [224 x 224 x frames] image series.
- The initial baseline used the direct 58-point z/partition IFFT.
- Later work addresses forward-map estimation, z-grid consistency, scaling, and conditioning.
- The slides describe the original stable path, not all later corrections.